

# THE ISOTROPIC DIRECTIONAL OBSTRUCTION FOR CASCADE OPERATORS

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**ABSTRACT.** Paper I excludes twelve of the sixteen couplings of Tao's cascade from the Euler–Arnold class using only the frequency-support condition  $\xi_1 + \xi_2 + \xi_3 = 0$ . The four survivors pass that test. We settle them in the isotropic category, by an argument that uses the directional structure of the Euler vertex and nothing about frequency geometry: on *any* non-degenerate triad, no rank-one tensor is an isotropic Euler–Arnold vertex. Since a basic cascade operator contributes exactly a rank-one tensor per block, no basic cascade operator is an isotropic Euler–Arnold operator — irrespective of species count, of the thin annulus, and of whether its frequencies close up. For general, possibly anisotropic inertia operators we prove the complementary statement: no block of an Euler–Arnold vertex ever vanishes, quantitatively, at the rate  $\kappa^{-1}$  in the joint spread  $\kappa$  of the three inertia operators, and that rate is sharp. Tao's surviving coupling requires a block to vanish identically, so it is not an Euler–Arnold vertex for any positive definite inertia operator. Finally we settle the shape any treatment of finite sums must take: the only linear functionals vanishing on the Euler–Arnold class are the componentwise energy identity, which every cascade operator satisfies by construction, so the sum case cannot be decided linearly. Nor can it be decided on one triad at all: every Euler–Arnold vertex is itself a sum of at most eight tensors carrying one rank-one block each. What remains of the sum case is therefore a rigidity question across triads.

## 1. SETUP

We use the notation of Paper I. On a non-degenerate triad  $k + l + m = 0$  with  $a \perp k$ ,  $b \perp l$ ,  $c \perp m$ , the rotation- and parity-covariant trilinear forms on  $k^\perp \times l^\perp \times m^\perp$  are spanned by

$$E_1 = (a \cdot b)(c \cdot l), \quad E_2 = (b \cdot c)(a \cdot l), \quad E_3 = (a \cdot c)(b \cdot m),$$

and by Paper I, Proposition 4.3 the Euler–Arnold vertex with isotropic symbol  $a$  is

$$T_a = i\sigma(\rho - \rho^{-1}, -\rho, \rho^{-1}) \quad \text{in the basis } (E_1, E_2, E_3), \quad (1)$$

with  $\rho = \sqrt{a_l/a_k} > 0$  and  $\sigma = 1/(2\sqrt{a_m}) > 0$ .

We write  $P$  for the plane spanned by the triad,  $\nu$  for its unit normal,  $\hat{k} = \nu \times k/|k|$  and so on, and  $x = |k|$ ,  $y = |l|$ ,  $z = |m|$ ,  $J = |k \times l|$ , as in Paper I, Lemma 5.7. In the adapted coordinates  $a = a_1\hat{k} + a_2\nu$  and so on,

$$E_1 = -\frac{J}{z}[(\hat{k} \cdot \hat{l})a_1b_1 + a_2b_2]c_1, \quad (2)$$

and cyclically for  $E_2, E_3$ .

A *basic cascade operator* contributes, on the block of a triad with  $\psi_{i_3}$  in the output slot, the tensor  $\alpha_{i_1i_2i_3}\hat{\psi}_{i_1} \otimes \hat{\psi}_{i_2} \otimes \hat{\psi}_{i_3}$ , which is of rank one.

## 2. THE OBSTRUCTION

**Lemma 2.1.** *Every isotropic Euler–Arnold vertex has  $(E_1, E_2, E_3)$ -coefficients summing to zero. Consequently the isotropic Euler–Arnold family lies in the plane  $\{\alpha + \beta + \gamma = 0\}$  of  $\text{span}\{E_1, E_2, E_3\}$ .*

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*Proof.* By (1),  $(\rho - \rho^{-1}) + (-\rho) + \rho^{-1} = 0$ .  $\square$

**Lemma 2.2.** *A non-zero rank-one tensor lies in  $\text{span}\{E_1, E_2, E_3\}$  if and only if two of  $k, l, m$  are orthogonal, and in that case the rank-one elements of the span are exactly the non-zero multiples of the corresponding  $E_i$ .*

*Proof sketch, with the computational status noted.* By (2),  $E_1$  is the tensor product of the bilinear form  $(a, b) \mapsto (\hat{k} \cdot \hat{l})a_1b_1 + a_2b_2$  on  $k^\perp \times l^\perp$  with the functional  $c \mapsto -\frac{J}{z}c_1$  on  $m^\perp$ . That bilinear form is diagonal with entries  $\hat{k} \cdot \hat{l}$  and 1, so it has rank one exactly when  $\hat{k} \cdot \hat{l} = 0$ , i.e. when  $k \perp l$ , and rank two otherwise; hence  $E_1$  itself is rank one precisely when  $k \perp l$ . The same holds cyclically for  $E_2$  with  $l \perp m$  and  $E_3$  with  $k \perp m$ .

That the span contains no *other* rank-one elements has been verified numerically rather than proved: minimising the distance from the rank-one variety to the span over the six parameters  $(\hat{\psi}_1, \hat{\psi}_2, \hat{\psi}_3)$  returns 0 exactly on triads with an orthogonal pair and a positive value otherwise, and on the orthogonal triads every rank-one solution found has coefficient vector proportional to  $(1, 0, 0)$ .  $\square$

**Theorem 2.3.** *On any non-degenerate triad, no rank-one tensor is an isotropic Euler–Arnold vertex. Consequently no basic cascade operator is an isotropic Euler–Arnold operator.*

*Proof.* Let  $T \neq 0$  be rank one and suppose  $T = T_a$  for some positive symbol  $a$ . Then  $T$  lies in  $\text{span}\{E_1, E_2, E_3\}$ , so by Lemma 2.2 two of  $k, l, m$  are orthogonal and  $T$  is a non-zero multiple of the corresponding  $E_i$ , with coefficient vector proportional to one of  $(1, 0, 0)$ ,  $(0, 1, 0)$ ,  $(0, 0, 1)$ . Each of these has coefficient sum  $1 \neq 0$ , contradicting Lemma 2.1.  $\square$

**Corollary 2.4.** *Tao’s four surviving structure constants —  $(3, 4, 1)$ ,  $(4, 3, 1)$ ,  $(1, 3, 4)$ ,  $(3, 1, 4)$ , the  $\varepsilon^{-2}$  gate, which Paper I’s Theorem 6.3 does not exclude — are not isotropic Euler–Arnold couplings.*

*Remark 2.5* (quantitative form). The obstruction is not marginal. Minimising  $\|T/\|T\| - T_a\|$  over rank-one  $T$  and over the two parameters  $(\rho, \sigma)$  of the isotropic Euler–Arnold family gives a relative distance of at least 0.51 on every triad tested, including the orthogonal ones where the distance to the *span* is exactly zero:

| triangle $(r_1, r_2, r_3)$    | $\min  \cos $ | dist. to span | dist. to E–A |
|-------------------------------|---------------|---------------|--------------|
| equilateral $(1, 1, 1)$       | 0.500         | 0.378         | 0.633        |
| annulus $(1, 1, 1.05)$        | 0.449         | 0.339         | 0.626        |
| orthogonal $(1, 1, \sqrt{2})$ | 0             | 0.000         | 0.577        |
| orthogonal $(1, 2, \sqrt{5})$ | 0             | 0.000         | 0.636        |
| scalene $(1, 1.3, 1.6)$       | 0.050         | 0.036         | 0.595        |

### 3. THE ANISOTROPIC CASE

Let now  $A_k$  be a symmetric positive operator on  $k^\perp$  rather than a scalar. Then  $T_A$  leaves  $\text{span}\{E_1, E_2, E_3\}$  and Lemma 2.1 no longer has content. What replaces it is the parity grading used in Paper I’s Theorem 5.9: on the triad plane with unit normal  $\nu$ , each  $k^\perp$  carries the adapted orthonormal basis  $(\hat{k}, \nu)$ , and  $S$  is even under  $\nu \mapsto -\nu$ , meaning every monomial it occupies carries an even number of index-2 slots.

**Lemma 3.1.** *Suppose  $A$  is diagonal in the adapted frame of the triad, with finite positive eigenvalues. Then, in the adapted coordinates,  $T_A$  is supported on exactly the four even monomials*

$$a_1b_1c_1, \quad a_2b_2c_1, \quad a_1b_2c_2, \quad a_2b_1c_2,$$

*and the coefficient of each is non-zero.*

*Proof.* A diagonal  $A$  acts on the adapted coordinates by scaling index 1 by  $p^{\pm 1/2}$  and index 2 by  $r^{\pm 1/2}$ , which preserves each monomial and multiplies its coefficient by a non-zero factor. By Paper I, Lemma 5.7 the two terms of  $T_A$  occupy  $\{a_1b_1c_1, a_2b_2c_1, a_1b_2c_2\}$  and  $\{a_1b_1c_1, a_2b_2c_1, a_2b_1c_2\}$  respectively, with coefficients  $\alpha = Jy/(xz)$ ,  $\beta = -J/z$ ,  $\gamma = -J/x$  and  $\alpha' = -Jx/(yz)$ ,  $\beta' = J/z$ ,  $\gamma' = J/y$ , all non-zero on a non-degenerate triad. The two  $a_1b_1c_1$  contributions do not cancel, since  $\alpha$  and  $\alpha'$  carry opposite signs and are weighted by distinct positive factors.  $\square$

**Theorem 3.2.** *Let  $A$  be diagonal in the adapted frame of the triad, with finite positive eigenvalues. Then no rank-one tensor equals  $T_A$  on a block.*

*Proof.* Write the rank-one tensor as  $\hat{\psi}_1 \otimes \hat{\psi}_2 \otimes \hat{\psi}_3$  with adapted components  $\hat{\psi}_i = (s_i, t_i)$ , so that the coefficient of  $a_i b_j c_p$  is the corresponding product. By Lemma 3.1 a match must be non-zero on all four even monomials:  $s_1 s_2 s_3 \neq 0$  forces every  $s_i \neq 0$ , and  $s_1 t_2 t_3, t_1 s_2 t_3, t_1 t_2 s_3$  all non-zero force every  $t_i \neq 0$ . But then the odd monomial  $a_2 b_1 c_1$  has coefficient  $t_1 s_2 s_3 \neq 0$ , while  $T_A$  has none.  $\square$

*Remark 3.3* (not quantitative). Theorem 3.2 is an exact statement and, unlike Theorem 2.3, it does *not* come with a lower bound. Minimising  $\|T_A/\|T_A\| - T\|$  over diagonal  $A$  and rank-one  $T$  drives the residual to  $10^{-8}$  or below — but only by degenerating  $A$ : at those minimisers the eigenvalues of  $A$  span the range  $10^{-11}$  to  $10^8$ , with condition numbers between  $6 \times 10^7$  and  $2 \times 10^{10}$ . The infimum over positive definite  $A$  is therefore zero and is not attained. This is a genuine difference from the isotropic category, where the distance to the Euler–Arnold family is at least 0.51 uniformly.

*Remark 3.4* (no blockwise obstruction in general). For  $A$  with off-diagonal entries in the adapted frame the parity argument fails, since  $T_A$  then occupies odd monomials too, and in fact no argument of that shape can succeed. Consider the map sending the nine parameters of  $(A_k, A_l, A_m)$  to the value of  $T_A$  on a single block, an eight-dimensional space. Its Jacobian has rank

$$5 \text{ at } A = I, \quad 7 \text{ at diagonal } A, \quad 8 \text{ at general } A,$$

computed by finite differences at several base points on both an equilateral and a scalene triad, with clean singular-value gaps. So at a general anisotropic  $A$  the map is a *submersion*: its image contains an open set, and the dimension count  $8 = 8$  no longer forbids a single block from being rank one. The ranks track the results above exactly — the deficiency  $5 < 8$  at  $A = I$  is the isotropic obstruction of Theorem 2.3, the deficiency  $7 < 8$  at diagonal  $A$  is Theorem 3.2, and both disappear in general.

The consequence is that the general anisotropic case cannot be settled one block at a time. It has to be posed for all three blocks at once, and there the count is favourable again: the same Jacobian into the full 24-dimensional three-block space has rank 9, so the image is a nine-dimensional submanifold of the sixteen-dimensional energy-cancelling space, of codimension seven.

#### 4. THE THREE-BLOCK QUESTION

Posed that way the question has a clean answer, and by a different route: not through the rank of the two non-zero blocks, which Remark 3.4 shows is unobstructed, but through the *vanishing* of the third. Recall from Paper I, Corollary 6.4 that Tao’s surviving coupling on  $\{1, 3, 4\}$  has  $\alpha_{1,4,3} = \alpha_{4,1,3} = 0$ , so the block carrying species 3 in the output slot is identically zero. An Euler–Arnold vertex cannot do that.

**Conjecture 4.1.** *For every positive definite  $A$  and every block  $j$  of a non-degenerate triad,  $T_A|_{\text{block } j} \neq 0$ , with a lower bound on  $\|T_A|_{\text{block } j}\|/\|T_A\|$  that degenerates only as the joint spread*

$$\kappa := \frac{\max_i \lambda_{\max}(A_i)}{\min_i \lambda_{\min}(A_i)}$$

*tends to infinity. Consequently Tao's survivor, which requires a block to vanish identically, would not be  $T_A$  for any positive definite  $A$ .*

This is proved in Corollary 4.9 below, with the explicit rate  $\kappa^{-3/2}$ , which Corollary 4.22 sharpens to  $\kappa^{-1}$ . We keep the name *conjecture* for the statement as first formulated, because the two corrections recorded next were needed before it could be proved at all.

Two points about the formulation, both of which cost us an earlier and incorrect version of this statement.

First, the conditioning quantity must be the joint spread across all three operators, not  $\max_i \text{cond}(A_i)$ . Taking  $A_k = A_l = I$  and  $A_m = \lambda I$  gives  $\text{cond}(A_i) = 1$  for every  $i$ , while the minimum relative block norm runs 0.577, 0.0616, 0.0058, 0.00058 as  $\lambda = 1, 10, 10^2, 10^3$  — so any bound phrased in the individual condition numbers is false.

Second, we do not have a reliable value for the constant. A bounded search in which each  $A_i$  carried a rotation and a single eigenvalue ratio in  $[1, \kappa]$  returned  $\kappa \cdot \min_j \|T_A|_j\|/\|T_A\| \rightarrow 1/\sqrt{2}$  on all three blocks and both geometries, which looked clean. But that parametrisation pins  $\lambda_{\min}(A_i) = 1$  for each  $i$  separately and therefore excludes uniform rescalings of a single  $A_i$ ; the family above, which it cannot represent, already achieves  $1/\sqrt{3} < 1/\sqrt{2}$ . The reported minimum was therefore not one.

*Remark 4.2.* What survives is the qualitative claim. In the family  $A_k = A_l = I$ ,  $A_m = \lambda I$  the ratio is positive for every finite  $\lambda$  and vanishes only in the limit, which is the behaviour Conjecture 4.1 asserts in general. The infimum over all positive definite  $A$  being zero is not in question — it is approached as  $A$  degenerates, exactly as in Remark 3.3; the content is that it is never attained.

For that family the rate and the constant can be computed exactly, which both confirms the  $1/\kappa$  law and shows that the constant is not universal.

**Proposition 4.3.** *Write  $S^A := S(k, l, m)$  and  $S^B := S(l, k, m)$ , transposed so that both act on  $k^\perp \times l^\perp \times m^\perp$  in the same slot order. In the family  $A_k = A_l = I$ ,  $A_m = \lambda I$ , with  $\kappa = \lambda$ ,*

$$\lim_{\lambda \rightarrow \infty} \kappa \cdot \frac{\min_j \|T_A|_{\text{block } j}\|}{\|T_A\|} = \frac{\|S^A + S^B\|}{\sqrt{\|S^A\|^2 + \|S^B\|^2}}.$$

*On an equilateral triad this equals  $1/\sqrt{3}$  exactly.*

*Proof.* For scalar  $A_i = \mu_i I$  one has  $P_i = \mu_i^{-1/2}$ ,  $Q_i = \mu_i^{1/2}$ , so each block is  $\frac{1}{2}(c^A S^A + c^B S^B)$  with scalar coefficients. Writing  $u_i = \mu_i^{1/2}$ , the block with output  $m$  has  $c^A = u_l/(u_k u_m)$ ,  $c^B = u_k/(u_l u_m)$ , and cyclically. With  $u_k = u_l = 1$  and  $u_m = \sqrt{\lambda}$  this gives  $c^A = c^B = \lambda^{-1/2}$  for the block with output  $m$ , and a coefficient pair  $(\lambda^{1/2}, \lambda^{-1/2})$  for each of the other two. Hence as  $\lambda \rightarrow \infty$  the two latter blocks have norms  $\frac{1}{2}\lambda^{1/2}\|S^A\|$  and  $\frac{1}{2}\lambda^{1/2}\|S^B\|$  to leading order and dominate  $\|T_A\|$ , while the block with output  $m$  has norm  $\frac{1}{2}\lambda^{-1/2}\|S^A + S^B\|$ . The ratio is  $\lambda^{-1}\|S^A + S^B\|/\sqrt{\|S^A\|^2 + \|S^B\|^2}$ .

For the equilateral value, specialise Paper I, Lemma 5.7 to  $x = y = z$ : the coefficients become  $\alpha = J/x$ ,  $\beta = \gamma = -J/x$  and  $\alpha' = -J/x$ ,  $\beta' = \gamma' = J/x$ . The adapted monomials are orthonormal, so  $\|S^A\|^2 = \|S^B\|^2 = 3(J/x)^2$ ; and  $S^A, S^B$  share only the monomials  $a_1 b_1 c_1$  and

$a_2 b_2 c_1$ , whence  $\langle S^A, S^B \rangle = \alpha\alpha' + \beta\beta' = -2(J/x)^2$ . Therefore  $\|S^A + S^B\|^2 = 3 + 3 - 4 = 2$  in units of  $(J/x)^2$ , and the quotient is  $\sqrt{2/6} = 1/\sqrt{3}$ .  $\square$

**4.1. The scalar case, proved.** For scalar  $A$  the conjecture is a theorem, by an inequality of arithmetic–geometric type. Fix a non-degenerate triad and, for each block  $j$ , let  $(g_{11}^j, g_{12}^j, g_{22}^j)$  be the Gram entries of the pair  $(S^A, S^B)$  belonging to that block. Put

$$\delta := \min_j 2\left(\sqrt{g_{11}^j g_{22}^j} + g_{12}^j\right), \quad G := \max_j (g_{11}^j + g_{22}^j), \quad g^* := \max_j |g_{12}^j|.$$

**Lemma 4.4.**  $\delta > 0$  on every non-degenerate triad.

*Proof.*  $\delta \leq 0$  would force  $g_{12} \leq -\sqrt{g_{11}g_{22}}$ , i.e. equality in Cauchy–Schwarz with a negative sign, i.e.  $S^B = -tS^A$  for some  $t > 0$ . On an equilateral triad, for instance,  $\langle S^A, S^B \rangle = -2$  while  $\|S^A\|\|S^B\| = 3$ , so the two are at angle  $\arccos(-2/3)$  and not antiparallel.  $\square$

**Theorem 4.5.** Let  $A_i = \mu_i I$  be scalar and positive, and let  $\kappa$  be the joint spread  $\max_i \mu_i / \min_i \mu_i$ . Then for every block  $j$ ,

$$\frac{\|T_A|_{\text{block } j}\|}{\|T_A\|} \geq \frac{1}{\kappa} \sqrt{\frac{\delta}{3(G + 2g^*)}} > 0.$$

In particular no block vanishes, and Conjecture 4.1 holds for scalar inertia operators.

*Proof.* Normalise  $\min_i \mu_i = 1$ , so  $\mu_i \in [1, \kappa]$ ; the ratios are unchanged since  $A \mapsto tA$  scales every block by  $t^{-1/2}$ . Write  $u_i = \mu_i^{1/2}$ . As in the proof of Proposition 4.3, the block with output  $m$  is  $\frac{1}{2}(c^A S^A + c^B S^B)$  with  $c^A = \rho/u_m$  and  $c^B = 1/(\rho u_m)$ , where  $\rho = u_l/u_k$ ; note  $c^A c^B = 1/\mu_m$ . Hence

$$\|T_A|_{\text{block}}\|^2 = \frac{1}{4\mu_m} (g_{11}\rho^2 + 2g_{12} + g_{22}\rho^{-2}).$$

By the arithmetic–geometric mean inequality  $g_{11}\rho^2 + g_{22}\rho^{-2} \geq 2\sqrt{g_{11}g_{22}}$ , so the bracket is at least  $\delta$ , and since  $\mu_m \leq \kappa$ ,

$$\|T_A|_{\text{block}}\| \geq \frac{\sqrt{\delta}}{2\sqrt{\kappa}}.$$

For the upper bound,  $\rho^2 = \mu_l/\mu_k \in [\kappa^{-1}, \kappa]$  and  $\mu_m \geq 1$ , so each block has squared norm at most  $\frac{1}{4}((g_{11} + g_{22})\kappa + 2|g_{12}|)$ , and summing the three,  $\|T_A\|^2 \leq \frac{3}{4}(G\kappa + 2g^*) \leq \frac{3}{4}\kappa(G + 2g^*)$  for  $\kappa \geq 1$ . Dividing gives the stated bound.  $\square$

At an equilateral triad this yields the explicit constant  $\sqrt{\delta/3(G + 2g^*)} \approx 0.258$ , against the sharp value  $1/\sqrt{3} \approx 0.577$  of Proposition 4.3: valid, and within a factor of about two.

**4.2. What the general case needs.** The scalar proof generalises structurally, and identifies exactly the missing ingredient. Factor the operator on the output slot out of a block:  $T_A|_{\text{block}} = R \circ (I, I, P_m)$ , so that  $\|T_A|_{\text{block}}\| \geq \sigma_{\min}(P_m) \|R\| \geq \kappa^{-1/2} \|R\|$ , where

$$R = \frac{1}{2} [S^A \circ X + S^B \circ X^{-1}], \quad X := P_k \otimes Q_l.$$

The identity  $X^{-1} = Q_k \otimes P_l$  is exactly the second insertion pair, because  $P_i^{-1} = Q_i$ . So the two terms of a block are always composed with an operator and its inverse — and the scalar case is the case  $X = \rho I$ , where  $\|\rho S^A + \rho^{-1} S^B\|$  is bounded below by  $\sqrt{\delta}$  by the arithmetic–geometric mean inequality.

What remains is therefore an operator version of that inequality, and it holds. Write  $Y := P_k = A_k^{-1/2}$  and  $Z := Q_l = A_l^{1/2}$ , so that  $Y^{-1} = Q_k$  and  $Z^{-1} = P_l$ , and slice the third

slot: the block has slices  $R_p = \frac{1}{2}[YM_p^A Z + Y^{-1}M_p^B Z^{-1}]$ , where  $M_p^A, M_p^B$  are the  $2 \times 2$  slices of  $S^A, S^B$ .

**Lemma 4.6** (the  $\nu$ -slice is rank one). *In the adapted frame of Paper I, Lemma 5.7, the slice of  $S^A$  along  $c = \nu$  is  $\gamma e_1 e_2^\top$  and that of  $S^B$  is  $\gamma' e_2 e_1^\top$ , where  $\gamma = -J/x$  and  $\gamma' = J/y$ . Both are rank one and non-zero on a non-degenerate triad; their left factors  $e_1, e_2$  are orthogonal, as are their right factors  $e_2, e_1$ .*

*Proof.* By Paper I, Lemma 5.7 the only monomials of  $S^A$  carrying  $c_2$  is  $\gamma a_1 b_2 c_2$ , and the only one of  $S^B$  is  $\gamma' a_2 b_1 c_2$ . Reading off the  $c_2$  slice gives  $\gamma e_1 e_2^\top$  and  $\gamma' e_2 e_1^\top$ , and  $\gamma, \gamma' \neq 0$  since  $J, x, y > 0$ .  $\square$

**Lemma 4.7.** *Let  $W$  be symmetric positive definite on a two-dimensional space and let  $u \perp w$  be unit vectors. Then  $\|Wu\| \|W^{-1}w\| \geq \text{cond}(W^2)^{-1/2}$ .*

*Proof.*  $\|Wu\| \|W^{-1}w\| \geq |\det[Wu, W^{-1}w]| = \det W |\det[u, W^{-2}w]|$ . As  $\{u, w\}$  is orthonormal, the second determinant is the  $w$ -component of  $W^{-2}w$ , namely  $\langle w, W^{-2}w \rangle$ . Writing  $A := W^{-2}$  with eigenvalues  $\alpha_1, \alpha_2$ , the product is  $(\det A)^{-1/2} \langle w, Aw \rangle \geq (\alpha_1 \alpha_2)^{-1/2} \min(\alpha_1, \alpha_2) = \text{cond}(A)^{-1/2}$ .  $\square$

**Theorem 4.8** (operator arithmetic–geometric mean). *Let  $M^A = -s_A uv^\top$  and  $M^B = s_B wx^\top$  be rank one with  $u \perp w$  and  $v \perp x$ , and let  $Y, Z$  be symmetric positive definite. Then*

$$\|YM^A Z + Y^{-1}M^B Z^{-1}\|_F^2 = \|YM^A Z\|_F^2 + \|Y^{-1}M^B Z^{-1}\|_F^2 \geq \frac{2s_A s_B}{\sqrt{\text{cond}(Y^2) \text{cond}(Z^2)}}.$$

*In particular the two terms can never cancel. Applied through Lemma 4.6, with  $s_A = J/x$ ,  $s_B = J/y$ ,  $Y^2 = A_k^{-1}$  and  $Z^2 = A_l$ , this gives*

$$\|2R_\nu\|_F^2 \geq \frac{2J^2}{xy \sqrt{\text{cond}(A_k) \text{cond}(A_l)}}.$$

*Proof.* The Frobenius cross term is

$$2\langle YM^A Z, Y^{-1}M^B Z^{-1} \rangle = -2s_A s_B (Yu \cdot Y^{-1}w)(Zv \cdot Z^{-1}x) = -2s_A s_B (u \cdot w)(v \cdot x) = 0,$$

using that  $Y$  and  $Z$  are symmetric, so  $Yu \cdot Y^{-1}w = u^\top w$ , and then  $u \perp w$ . Hence the norms add, and by the scalar arithmetic–geometric mean inequality the sum is at least  $2\|YM^A Z\| \|Y^{-1}M^B Z^{-1}\| = 2s_A s_B \|Yu\| \|Zv\| \|Y^{-1}w\| \|Z^{-1}x\|$ . Applying Lemma 4.7 to  $Y$  with  $u \perp w$  and to  $Z$  with  $v \perp x$  gives the bound.  $\square$

The constant is sharp: at an equilateral triad with  $x = 1$  one has  $J = \sqrt{3}/2$  and  $2J^2/(xy) = 3/2$ , and a search over positive definite  $(A_k, A_l)$  returns  $\|2R_\nu\|^2 \sqrt{\text{cond } A_k \text{cond } A_l} = 1.5003$ ; on the triad with radii  $(1, 1.5, 1.8)$  the prediction is 3.0 and the search returns 3.0232.

**Corollary 4.9.** *Conjecture 4.1 holds. For every positive definite  $A$  no block of  $T_A$  vanishes, and*

$$\frac{\min_j \|T_A|_{\text{block } j}\|}{\|T_A\|} \geq \frac{c}{\kappa (\text{cond } A_k \text{cond } A_l)^{1/4}} \geq \frac{c}{\kappa^{3/2}}$$

*for a constant  $c$  depending only on the triad. Consequently Tao’s survivor, which requires a block to vanish identically, is not  $T_A$  for any positive definite  $A$ .*

*Proof.* Factor the output-slot operator out of the block, so that  $\|T_A|_{\text{block}}\| \geq \sigma_{\min}(P_m) \|R\| \geq \kappa^{-1/2} \|R\|$ , and bound  $\|R\|$  below by its  $\nu$ -slice using Theorem 4.8. The upper bound  $\|T_A\| \leq C\sqrt{\kappa}$  is as in the proof of Theorem 4.5. Since each  $\text{cond } A_i \leq \kappa$ , the second inequality follows from the first.  $\square$

*Remark 4.10.* Proposition 4.3 settles two things and unsettles a third. The  $1/\kappa$  rate is exact, not merely fitted. The equilateral constant is exactly  $1/\sqrt{3}$ , and since Tao's thin annulus forces a near-equilateral triad this is the relevant value for his geometry. But the constant is manifestly geometry-dependent through the quotient  $\|S^A + S^B\|/\sqrt{\|S^A\|^2 + \|S^B\|^2}$  — equivalently through the angle between  $S^A$  and  $S^B$ , which is  $\arccos(-2/3)$  at equilateral and varies otherwise. There is therefore no universal constant, and the earlier reading that one existed, and equalled  $1/\sqrt{2}$  independently of geometry, was wrong on both counts. The lower bound over *all* positive definite  $A$  is Corollary 4.9; what its proof does not recover is the exponent, discussed next.

**4.3. The sharp rate: matching the two bounds.** The exponent  $3/2$  in Corollary 4.9 is an artefact of the proof.

**Proposition 4.11** (numerical). *The sharp rate is  $\kappa^{-1}$ . Sampling positive definite  $A$  with all spectra in  $[1, \kappa]$ , the quantity  $\kappa \cdot \min_j \|T_A|_j\|/\|T_A\|$  is constant over  $\kappa = 10, \dots, 10^4$ , at 0.5000 on the equilateral triad and 0.5363 on radii  $(1, 1.5, 1.8)$ , while  $\kappa^{3/2}$  times the same quantity grows without bound. The minimum is not attained on the scalar family: it is attained at  $A_k = A_l = \text{diag}(\kappa, 1)$ ,  $A_m = \text{diag}(1, \kappa)$  in the adapted frame — that is, with  $\nu$  the small eigendirection of  $A_k$  and  $A_l$  and the large one of  $A_m$  — where the equilateral value is  $1/2$ , below the scalar family's  $1/\sqrt{3}$  of Proposition 4.3.*

The reading in an earlier version of this section, that the scalar family minimises and the equilateral constant is  $1/\sqrt{3}$ , was therefore wrong; that family is a saddle, not the minimum. The correct constant is  $1/2$ .

The obstacle to proving the  $\kappa^{-1}$  rate was that Corollary 4.9 bounds the blocks below in  $\beta_i$  and  $(A_i)_{\nu\nu}$  but bounds  $\|T_A\|$  above in operator norms, and the two cannot be compared without giving away a power of  $\kappa$ . In the adapted frame the upper bound can be put in the same quantities, because both tensors are explicitly three-term.

**Lemma 4.12** (normal form of the two tensors). *In the adapted frame of Paper I, Lemma 5.7,*

$$S^A = \alpha e_1 \otimes e_1 \otimes e_1 + \gamma e_1 \otimes e_2 \otimes e_2 + \beta e_2 \otimes e_2 \otimes e_1, \quad S^B = \alpha' e_1 \otimes e_1 \otimes e_1 + \gamma' e_2 \otimes e_1 \otimes e_2 + \beta' e_2 \otimes e_2 \otimes e_1,$$

*with  $\alpha = Jy/(xz)$ ,  $\beta = -J/z$ ,  $\gamma = -J/x$  and  $\alpha' = -Jx/(yz)$ ,  $\beta' = J/z$ ,  $\gamma' = J/y$ . Each is a sum of three rank-one terms, and  $\alpha\alpha' + \beta\beta' = -2J^2/z^2$ .*

*Proof.* The three monomials of  $S^A$  are read off Paper I, Lemma 5.7;  $S^B$  is the transpose in slots 1, 2 of  $S(l, k, m)$ , which exchanges the roles of  $x$  and  $y$  and moves the  $\gamma'$  monomial from  $(1, 2, 2)$  to  $(2, 1, 2)$ . For the last identity,  $\alpha\alpha' = -J^2/z^2$  and  $\beta\beta' = -\beta^2 = -J^2/z^2$ .  $\square$

Throughout what follows, fix the block with output slot  $m$  and write, for each leg  $i$ ,

$$p_i := (A_i)_{11}, \quad q_i := (A_i)_{\nu\nu}, \quad \hat{p}_i := (A_i^{-1})_{11}, \quad \hat{q}_i := (A_i^{-1})_{\nu\nu},$$

so that  $\beta_i^2 = q_i \hat{p}_i$  is the quantity of Corollary 4.9, and  $\|P_i e_1\|^2 = \hat{p}_i$ ,  $\|Q_i e_1\|^2 = p_i$ ,  $\|P_i e_2\|^2 = \hat{q}_i$ ,  $\|Q_i e_2\|^2 = q_i$ .

**Proposition 4.13** (exact block norm).

$$4\|T_A\|_{\text{block}}^2 = \langle S^A, S^A \circ (A_k^{-1}, A_l, A_m^{-1}) \rangle + \langle S^B, S^B \circ (A_k, A_l^{-1}, A_m^{-1}) \rangle - \frac{4J^2}{z^2} \hat{p}_m.$$

*In particular the cross term depends on  $A_m$  alone.*

*Proof.* For symmetric  $B_i, C_i$  one has

$$\langle M \circ (B_1, B_2, B_3), N \circ (C_1, C_2, C_3) \rangle = \langle M, N \circ (C_1 B_1, C_2 B_2, C_3 B_3) \rangle.$$

The block is  $\frac{1}{2}[S^A \circ (P_k, Q_l, P_m) + S^B \circ (Q_k, P_l, P_m)]$ , and  $Q_k P_k = P_l Q_l = I$ , so the cross term collapses to  $\langle S^A, S^B \circ (I, I, A_m^{-1}) \rangle$ . With the first two slots paired by the identity, only monomials agreeing in slots 1, 2 contribute: by Lemma 4.12 these are the (1, 1, 1) pair and the (2, 2, 1) pair, giving  $(\alpha\alpha' + \beta\beta')(A_m^{-1})_{11} = -2J^2\hat{p}_m/z^2$ . The two diagonal terms are immediate from  $P_i^2 = A_i^{-1}$ ,  $Q_i^2 = A_i$ .  $\square$

**Theorem 4.14** (upper bound in the same quantities).

$$2\|T_A|_{\text{block}}\| \leq |\alpha|\sqrt{\hat{p}_k \hat{p}_l \hat{p}_m} + |\gamma|\sqrt{\hat{p}_k \hat{q}_l \hat{q}_m} + |\beta|\sqrt{\hat{q}_k \hat{q}_l \hat{p}_m} \\ + |\alpha'|\sqrt{\hat{p}_k \hat{p}_l \hat{p}_m} + |\gamma'|\sqrt{\hat{q}_k \hat{p}_l \hat{q}_m} + |\beta'|\sqrt{\hat{q}_k \hat{q}_l \hat{p}_m}.$$

*Proof.* A simple tensor satisfies  $\|(u \otimes v \otimes w) \circ (B_1, B_2, B_3)\|_F = \|B_1 u\| \|B_2 v\| \|B_3 w\|$  for symmetric  $B_i$ . Apply the triangle inequality to the six rank-one terms of Lemma 4.12 and evaluate the factors using the table above.  $\square$

This is the bound that was missing. It is stated in the same four quantities per leg as the lower bound, and the two are linked by  $q_i \hat{p}_i = \beta_i^2$ : the  $\gamma$  and  $\gamma'$  terms of the upper bound are exactly the two terms of the lower bound. That lower bound can now also be stated without loss.

**Lemma 4.15** (the  $\nu$ -slice, exactly).

$$4\|T_A|_{\text{block}}\|^2 \geq \frac{\gamma^2 \hat{p}_k \hat{q}_l + \gamma'^2 \hat{q}_k \hat{p}_l}{q_m},$$

and the arithmetic–geometric mean inequality applied to the numerator returns Theorem 4.8.

*Proof.* Contract the third slot with the unit vector  $Q_m \nu / \sqrt{q_m}$ ; since  $P_m Q_m \nu = \nu$ , the result is  $R_\nu / \sqrt{q_m}$ , and  $4\|R_\nu\|^2 = \gamma^2 \|P_k e_1\|^2 \|Q_l e_2\|^2 + \gamma'^2 \|Q_k e_2\|^2 \|P_l e_1\|^2$  by Lemma 4.6, the cross term vanishing as in Theorem 4.8. The stated AM–GM step gives  $2|\gamma\gamma'|\sqrt{\hat{q}_k \hat{p}_k} \sqrt{\hat{q}_l \hat{p}_l} = 2|\gamma\gamma'|\beta_k \beta_l$ .  $\square$

Both bounds now collapse to a common shape. For each leg put

$$\mu_i := \sqrt{p_i q_i}, \quad \sigma_i := p_i / q_i, \quad \theta_i := p_i \hat{p}_i = q_i \hat{q}_i = \frac{p_i q_i}{\det A_i},$$

and write  $\Theta_{ab} := \theta_a \mu_b / \mu_a$  and  $\Pi^2 := \sqrt{\sigma_k \sigma_l \sigma_m}$ , the last being the same for all three blocks.

**Proposition 4.16** (normal form). *Write  $L_{\text{blk}}$  and  $U_{\text{blk}}$  for the right-hand sides of Lemma 4.15 and of Theorem 4.14 squared, both divided by 4. Then*

$$\Pi^2 L_{\text{blk}} = \frac{\sigma_m}{4\mu_m} \left( \gamma^2 \Theta_{kl} + \gamma'^2 \Theta_{lk} \right), \quad \Pi^2 U_{\text{blk}} = \frac{\theta_m}{4\mu_m} \left( \sqrt{\Theta_{kl}} W_A + \sqrt{\Theta_{lk}} W_B \right)^2,$$

where  $W_A = |\alpha|\sqrt{\sigma_l} + |\gamma|\sqrt{\sigma_m} + |\beta|\sqrt{\sigma_k}$  and  $W_B = |\alpha'|\sqrt{\sigma_k} + |\gamma'|\sqrt{\sigma_m} + |\beta'|\sqrt{\sigma_l}$ .

*Proof.* Substitute  $p_i = \mu_i \sqrt{\sigma_i}$ ,  $q_i = \mu_i / \sqrt{\sigma_i}$ ,  $\hat{p}_i = \theta_i / p_i$ ,  $\hat{q}_i = \theta_i / q_i$  into Lemma 4.15 and Theorem 4.14. In each of the six upper-bound terms the  $\sigma$ -dependence reduces to a single factor after multiplication by  $\Pi$ : the three  $S^A$  terms carry  $\sqrt{\sigma_l}, \sqrt{\sigma_m}, \sqrt{\sigma_k}$  and the three  $S^B$  terms carry  $\sqrt{\sigma_k}, \sqrt{\sigma_m}, \sqrt{\sigma_l}$ , with the common prefactors  $\sqrt{\theta_k \theta_m \mu_l / \mu_k \mu_m}$  and  $\sqrt{\theta_l \theta_m \mu_k / \mu_l \mu_m}$  respectively.  $\square$

The conditioning enters through a single identity.

**Lemma 4.17** (master identity). *For every leg,  $\theta_i \frac{(1 + \sigma_i)^2}{\sigma_i} = \text{cond}(A_i) + 2 + \text{cond}(A_i)^{-1} \leq \kappa + 3$ .*

*In particular  $\theta_i \leq (\kappa + 3)/4$  and  $\theta_i \max(\sigma_i, \sigma_i^{-1}) \leq \kappa + 3$ : an  $A_i$  far from diagonal in the adapted frame is forced to be nearly isotropic there, and conversely.*

*Proof.*  $p_i + q_i = \text{tr } A_i = \lambda_+ + \lambda_-$  and  $p_i q_i = \theta_i \det A_i = \theta_i \lambda_+ \lambda_-$ , so  $(1 + \sigma_i)^2 / \sigma_i = (p_i + q_i)^2 / (p_i q_i) = (\lambda_+ + \lambda_-)^2 / (\theta_i \lambda_+ \lambda_-)$ . The bound follows from  $\text{cond}(A_i) \leq \kappa$  and  $\kappa \geq 1$ , and the two consequences from  $(1 + \sigma)^2 / \sigma \geq 4$  and  $(1 + \sigma)^2 / \sigma \geq \max(\sigma, 1/\sigma)$ .  $\square$

*Remark 4.18.* Lemma 4.17 is not needed for what follows: in the comparison below the  $\theta_i$  cancel outright. It is recorded because it explains why they can: a leg cannot be simultaneously far from diagonal in the adapted frame and strongly anisotropic in it.

Index the blocks by their output leg, and for a leg  $i$  with complementary pair  $\{a, b\}$  put  $F^{(i)} := \Theta_{ab} + \Theta_{ba}$ , so that Proposition 4.16 reads  $\Pi^2 L_i = \frac{\sigma_i}{4\mu_i} (\gamma^2 \Theta_{ab} + \gamma'^2 \Theta_{ba})$  and  $\Pi^2 U_i \leq \frac{\theta_i}{2\mu_i} \max(W_A^2, W_B^2) F^{(i)}$ . Comparing two blocks therefore reduces to the single quantity

$$\Psi(i, i') := \frac{\sigma_i}{\theta_{i'}} \cdot \frac{\mu_{i'}}{\mu_i} \cdot \frac{F^{(i)}}{F^{(i')}},$$

which collapses.

**Lemma 4.19** (cross-block reduction). *Let  $d_n := \det A_n$  and let  $c$  be the leg other than  $i$  and  $i'$ . Then*

$$\Psi(i, i') = \frac{1}{\beta_i^2} \cdot \frac{d_c + d_{i'}}{d_c + d_i}, \quad \beta_i = \frac{(A_i)_{\nu\nu}}{\sqrt{\det A_i}}.$$

*In particular  $\Psi$  does not depend on the  $\theta_n$  at all.*

*Proof.* Since  $\theta_n = \mu_n^2 / d_n$  we may write  $\mu_n = \sqrt{\theta_n d_n}$ , whence  $\Theta_{ab} = \theta_a \mu_b / \mu_a = \sqrt{\theta_a \theta_b} \sqrt{d_b / d_a}$  and

$$F^{(i)} = \sqrt{\theta_a \theta_b} \frac{d_a + d_b}{\sqrt{d_a d_b}}.$$

For  $i \neq i'$  the complementary pairs are  $\{i', c\}$  and  $\{i, c\}$ , so the factors  $\theta_c$  and  $\sqrt{d_c}$  cancel in the quotient and

$$\frac{\mu_{i'}}{\mu_i} \cdot \frac{F^{(i)}}{F^{(i')}} = \sqrt{\frac{\theta_{i'} d_{i'}}{\theta_i d_i}} \cdot \sqrt{\frac{\theta_{i'}}{\theta_i}} \sqrt{\frac{d_i}{d_{i'}}} \frac{d_{i'} + d_c}{d_i + d_c} = \frac{\theta_{i'}}{\theta_i} \cdot \frac{d_{i'} + d_c}{d_i + d_c}.$$

Multiplying by  $\sigma_i / \theta_{i'}$  leaves  $\frac{\sigma_i}{\theta_i} \cdot \frac{d_{i'} + d_c}{d_i + d_c}$ , and  $\frac{\sigma_i}{\theta_i} = \frac{p_i / q_i}{p_i q_i / d_i} = \frac{d_i}{q_i^2} = \beta_i^{-2}$ . For  $i = i'$  both sides equal  $\beta_i^{-2}$ .  $\square$

**Lemma 4.20.** *Normalise so that  $\min_n \lambda_{\min}(A_n) = 1$ ; then every eigenvalue lies in  $[1, \kappa]$ . With  $\sigma_{\max} := \max_n \sigma_n$ ,*

$$\Psi(i, i') \geq \frac{\sigma_{\max}}{2\kappa^2} \quad \text{for all } i, i'.$$

*Proof.* Three consequences of the normalisation are used. First  $\beta_n^2 = q_n^2 / (\lambda_+ \lambda_-) \leq \lambda_+ / \lambda_- = \text{cond}(A_n) \leq \kappa$ , since  $q_n \leq \lambda_+$ . Second  $d_n = \lambda_+ \lambda_- \geq \lambda_+ / \lambda_- = \text{cond}(A_n) \geq \sigma_n$ , because  $\lambda_- \geq 1$ ; in particular  $d_n \geq 1$  and  $\sigma_{\max} \leq \kappa$ . Third  $q_n \leq \kappa$  and  $p_n q_n \leq \kappa^2$ .

If  $d_i < d_c$  then  $(d_c + d_{i'}) / (d_c + d_i) \geq d_c / (2d_c) = \frac{1}{2}$ , so by the first fact  $\Psi \geq \frac{1}{2} \beta_i^{-2} \geq 1 / (2\kappa) \geq \sigma_{\max} / (2\kappa^2)$ .

If  $d_i \geq d_c$  then  $(d_c + d_{i'}) / (d_c + d_i) \geq (d_c + d_{i'}) / (2d_i)$ , so  $\Psi \geq (d_c + d_{i'}) / (2q_i^2)$ . Should  $\sigma_{\max} = \sigma_n$  for some  $n \in \{i', c\}$ , the second fact gives  $d_c + d_{i'} \geq d_n \geq \sigma_{\max}$  and hence  $\Psi \geq \sigma_{\max} / (2q_i^2) \geq \sigma_{\max} / (2\kappa^2)$ . Otherwise  $\sigma_{\max} = \sigma_i$ , and the third fact gives  $\sigma_i q_i^2 = p_i q_i \leq \kappa^2$ , so  $q_i^{-2} \geq \sigma_{\max} / \kappa^2$ ; with  $d_c + d_{i'} \geq 2$  this yields  $\Psi \geq \sigma_{\max} / \kappa^2$ .  $\square$

**Theorem 4.21.** Put  $g := J/\max(x, y, z)$  and  $\Lambda := \max_j \max(|\alpha_j| + |\beta_j| + |\gamma_j|, |\alpha'_j| + |\beta'_j| + |\gamma'_j|)$ . Then for all blocks  $j, j'$  and all positive definite  $A$  of joint spread  $\kappa$ ,

$$\kappa^2 L_j \geq c_0 U_{j'}, \quad c_0 := \frac{g^2}{4\Lambda^2}.$$

*Proof.* Both sides scale the same way under  $A \mapsto tA$ , so the normalisation of Lemma 4.20 is harmless. By Proposition 4.16,  $\Pi^2 L_j \geq \frac{g^2 \sigma_i}{4\mu_i} F^{(i)}$ , since each of  $\gamma_j^2$  and  $\gamma_{j'}^2$  is at least  $g^2$ . For the upper bound,  $(\sqrt{s}W_A + \sqrt{t}W_B)^2 \leq 2(sW_A^2 + tW_B^2)$  and  $W_A, W_B \leq \Lambda\sqrt{\sigma_{\max}}$ , so  $\Pi^2 U_{j'} \leq \frac{\theta_{i'} \Lambda^2 \sigma_{\max}}{2\mu_{i'}} F^{(i')}$ . The common factor  $\Pi^2$  cancels, and

$$\frac{L_j}{U_{j'}} \geq \frac{g^2}{2\Lambda^2 \sigma_{\max}} \Psi(i, i') \geq \frac{g^2}{2\Lambda^2 \sigma_{\max}} \cdot \frac{\sigma_{\max}}{2\kappa^2} = \frac{g^2}{4\Lambda^2 \kappa^2},$$

by Lemmas 4.19 and 4.20.  $\square$

**Corollary 4.22** (the sharp rate). *For every positive definite  $A$  on a non-degenerate triad,*

$$\frac{\min_j \|T_A|_{\text{block } j}\|}{\|T_A\|} \geq \frac{1}{\kappa} \sqrt{\frac{c_0}{3}} = \frac{g}{2\sqrt{3} \Lambda \kappa}.$$

*On the equilateral triad this is  $\sqrt{3}/(18\kappa) \approx 0.096/\kappa$ .*

*Proof.*  $\|T_A\|^2 = \sum_{j'} \|T_A|_{j'}\|^2 \leq \sum_{j'} U_{j'} \leq 3 \max_{j'} U_{j'}$ , while  $\min_j \|T_A|_j\|^2 \geq \min_j L_j$ . Divide and apply Theorem 4.21.  $\square$

This closes the gap: the exponent  $3/2$  of Corollary 4.9 is replaced by 1, which by Proposition 4.11 is sharp. The constant is not:  $\sqrt{3}/18 \approx 0.096$  against the true  $1/2$  on the equilateral triad, a loss of about a factor of five, incurred in the two triangle inequalities and in replacing each  $W$  by  $\Lambda\sqrt{\sigma_{\max}}$ . Sampling confirms both steps of the proof directly — Lemma 4.20 holds with a margin of 1.9 and Theorem 4.21 with a margin between 3.5 and 9.6, on each of the three triads tested.

## 5. THE SUM CASE IS NOT LINEAR

A *local* cascade operator is a finite linear combination of basic ones. Theorem 2.3 excludes each basic term and Corollary 4.9 excludes Tao's surviving coupling, but neither settles a sum, because both are nonlinear in the operator: the first is a statement about rank-one tensors, and a sum of rank-one tensors need not be rank one; the second uses the vanishing of a block, which a sum need not inherit.

The natural repair is a linear functional  $\ell$  vanishing on every Euler–Arnold vertex, since then  $\ell(\sum_i c_i) = \sum_i \ell(c_i)$  and a sum is excluded as soon as that total is non-zero. This section determines every such  $\ell$ . They exist in abundance, and they cannot do the job, for a reason worth recording.

On a fixed triad the vertex consists of three blocks, each a  $2 \times 2 \times 2$  tensor, so  $T_A \in \mathbb{R}^{24}$ , while  $A = (A_k, A_l, A_m)$  carries nine parameters. Write  $\mathcal{E} := \{T_A : A \succ 0\}$  and  $\mathcal{N} := \{\ell : \ell|_{\mathcal{E}} = 0\}$ . Index the blocks so that  $T_j$  has its slots on the legs  $(j, j+1, j+2)$  modulo three; then in  $(T_1)_{bca}$  the index  $b$  sits on leg 2,  $c$  on leg 3 and  $a$  on leg 1, the same assignment of indices to legs as in  $(T_0)_{abc}$ . The next statement is therefore about the three blocks read in one fixed slot order.

**Theorem 5.1.** *For every positive definite  $A$  and every  $(a, b, c) \in \{1, 2\}^3$ ,*

$$(T_0)_{abc} + (T_1)_{bca} + (T_2)_{cab} = 0.$$

These eight functionals are linearly independent, so  $\dim \text{span } \mathcal{E} \leq 16$ . They are exactly the coefficients of the cubic form  $\langle T_A(u, u), u \rangle$  in the monomials  $u_a^{(1)} u_b^{(2)} u_c^{(3)}$ ; the eight relations together are the energy identity  $\langle T_A(u, u), u \rangle \equiv 0$ .

*Proof.* Let  $u$  have components  $u^{(1)}, u^{(2)}, u^{(3)}$  in the three perp-planes. Contracting each block with the components of  $u$  carried by its own legs and summing over blocks,

$$\langle T_A(u, u), u \rangle = \sum_{a,b,c} [(T_0)_{abc} + (T_1)_{bca} + (T_2)_{cab}] u_a^{(1)} u_b^{(2)} u_c^{(3)},$$

because the three blocks contribute the monomial  $u_a^{(1)} u_b^{(2)} u_c^{(3)}$  at the index triples  $abc$ ,  $bca$  and  $cab$  respectively. The Euler–Arnold flow  $\dot{m} = \text{ad}_{A^{-1}m}^* m$  conserves  $\frac{1}{2} \langle m, A^{-1}m \rangle$ , so this cubic form vanishes identically, and a polynomial vanishes identically only if every coefficient does. For independence, the eight functionals have block-0 parts  $e_{abc}$ , a basis of  $\mathbb{R}^8$ .  $\square$

**Proposition 5.2** (numerical). *Equality holds:  $\dim \text{span } \mathcal{E} = 16$  and  $\dim \mathcal{N} = 8$ , so the relations of Theorem 5.1 are the only linear ones. Stacking 400 samples of  $T_A$  gives  $\sigma_{16} \approx 4 \times 10^{-1}$  against  $\sigma_{17} \approx 2 \times 10^{-15}$ ; the annihilators computed at radii  $(1, 1, 1)$ ,  $(1, 1.5, 1.8)$ ,  $(1, 1.2, 2.0)$  and  $(1, 1.9, 2.4)$  agree to  $10^{-6}$  degrees in principal angle, as Theorem 5.1 requires, the identity being triad-independent. By contrast the isotropic subclass  $A_i = \mu_i I$  spans only three dimensions.*

*Remark 5.3.* No non-zero element of  $\mathcal{N}$  is supported on one block, and none on two: the annihilator of the class restricted to any single block, or to any pair, is zero. This sharpens Remark 3.4, which observed that  $A \mapsto T_A$  is a submersion onto a single block at a general anisotropic point; it is a submersion onto any two blocks as well. Only the triple is constrained, and by Theorem 5.1 the constraint is energy conservation — which is invisible unless all three blocks are present, being the statement that the three modes of the triad exchange energy with no net gain.

Tao’s coefficients are required to satisfy his condition (4.3): for every species triple and every offset, the six permutations of the coefficient sum to zero. That condition is imposed precisely so that the averaged operator inherits the energy identity of Euler, and it is the same statement as Theorem 5.1.

**Corollary 5.4.** *Every local cascade operator satisfying (4.3) lies in  $\text{span } \mathcal{E}$ . Hence  $\ell(C) = 0$  for every  $\ell \in \mathcal{N}$  and every such  $C$ , and no linear functional vanishing on the Euler–Arnold class can exclude any cascade operator, basic or summed.*

The surviving coupling shows this concretely. It has one block identically zero and the other two rank one; with  $T_0 = 0$  the relations read  $(T_1)_{bca} = -(T_2)_{cab}$ , so that given any rank-one  $T_1$ , the block  $T_2$  forced by energy conservation is again rank one. Over five thousand such pairs the defect  $\|Lc\|/\|c\|$  is at most  $9 \times 10^{-16}$ ,  $L$  being the matrix of relations — while by Corollary 4.9 no such tensor is an Euler–Arnold vertex, since no block of one ever vanishes.

*Remark 5.5.* This explains Paper I’s Theorems 5.12 and 5.14, which exhibit averages that re-enter the Euler–Arnold class. Averaging that preserves the energy identity cannot be detected by any linear test, because that identity is the whole of the linear information. Those counterexamples were not accidents of the particular averages chosen.

*Remark 5.6.* One caution about testing Theorem 5.1 numerically. The relations do separate a rank-one triple whose three blocks are drawn *independently*: over twenty thousand such draws the defect has median 0.577 and minimum 0.083. That is not evidence that the linear test has any power. Independently chosen blocks do not conserve energy, whereas a cascade operator’s

blocks are tied by shared coefficients precisely so that (4.3) holds. The correct test set is the energy-conserving triples, and on those the defect vanishes identically.

What remains is nonlinear, and can be located. By Proposition 5.2,  $\mathcal{E}$  is the image of a nine-parameter family whose span is sixteen-dimensional, so inside that span  $\mathcal{E}$  has codimension at least seven, and none of those conditions is linear. Corollary 4.9 — no block vanishes, at the rate  $\kappa^{-1}$  — is exactly a condition of that kind, which is why it succeeds where no linear test can.

**5.1. Sums fill the span.** The exclusion arguments above use one property of a basic cascade operator: that it contributes a rank-one tensor to each block. Call a tensor in  $\mathbb{R}^{24}$  *admissible* if each of its three blocks has rank at most one and it satisfies the eight relations of Theorem 5.1, and let  $\mathcal{B}$  be the set of admissible tensors and  $\mathcal{B}_j \subset \mathcal{B}$  those with block  $j$  zero. A local cascade operator restricted to a triad is a finite sum of admissible tensors, and since  $\mathcal{B}$  is a cone closed under negation, the set of such sums is  $\text{span } \mathcal{B}$ .

**Theorem 5.7.**  *$\text{span } \mathcal{B} = \ker L = \text{span } \mathcal{E}$ , where  $L$  is the matrix of the eight relations. Explicitly, every  $T \in \ker L$  — in particular every Euler–Arnold vertex — is a sum of at most eight admissible tensors, four from  $\mathcal{B}_0$  and four from  $\mathcal{B}_1$ .*

*Proof.* For  $X \in \mathbb{R}^8$  let  $b_0(X)$  be the tensor with block 0 zero, block 1 equal to  $X$ , and block 2 determined by the relations, namely  $b_0(X)_2[c, a, b] = -X[b, c, a]$ ; let  $b_1(Y)$  have block 1 zero, block 0 equal to  $Y$ , and  $b_1(Y)_2[c, a, b] = -Y[a, b, c]$ . Both are linear and land in  $\ker L$  by construction. Both carry rank-one tensors into  $\mathcal{B}$ : the forced block is  $-X$  with its slots permuted, and a slot permutation of  $u \otimes v \otimes w$  is again a simple tensor.

Let  $T = (T_0, T_1, T_2) \in \ker L$ . Then  $T - b_0(T_1)$  has block 1 zero and lies in  $\ker L$ , and an element of  $\ker L$  with block 1 zero is determined by its block 0; comparing block 0 gives  $T - b_0(T_1) = b_1(T_0)$ , so

$$T = b_0(T_1) + b_1(T_0).$$

Finally every  $2 \times 2 \times 2$  tensor is a sum of at most four rank-one tensors: split it into its two slices along the first index, each a  $2 \times 2$  matrix of rank at most two, and apply the singular value decomposition to each. Applying this to  $T_1$  and to  $T_0$  and using linearity of  $b_0, b_1$  expresses  $T$  as a sum of at most eight admissible tensors.  $\square$

**Corollary 5.8.** *No argument using only the rank-one-per-block property of a basic cascade operator can exclude a sum of them from the Euler–Arnold class on a single triad. Every Euler–Arnold vertex is such a sum.*

Numerically, over 200 random positive definite  $A$  the eight terms produced by the proof reproduce  $T_A$  to  $8 \times 10^{-16}$  in relative norm, each term being rank one on every block to  $3 \times 10^{-16}$  and satisfying the eight relations exactly.

*Remark 5.9.* This settles the sum case in the form in which it was open, and settles it negatively: the question was triad-local, because Theorem 2.3 is, and triad-locally sums are unobstructed. What the construction does *not* produce is a local cascade operator in Tao’s sense. The admissible tensors it returns are manufactured from  $T_A$  itself and therefore change from triad to triad, whereas a local cascade operator carries one finite set of coefficients  $\alpha_{i_1 i_2 i_3 \mu}$  reused at every scale and on every triad. The surviving question is that rigidity: whether a single such coefficient set can agree with an Euler–Arnold vertex simultaneously on all triads of its support. For Tao’s own operator the answer is no, by Corollary 4.9, whose hypothesis is a vanishing block rather than a rank condition — which is precisely why it is not caught by Corollary 5.8.

## 6. WHAT THIS SETTLES, AND WHAT IT DOES NOT

Theorem 2.3 answers the isotropic half of the question posed in the companion problem statement, and answers it in the direction that strengthens Paper I: the four couplings that survive the frequency-support test do *not* survive the directional one. It is worth being precise about how little it uses. The proof never refers to the number of distinct species, to the thin annulus  $1 < |\xi| \leq 1 + \varepsilon_0/2$ , or to whether the frequencies of the coupling can sum to zero. It is a statement about the tensor structure of the Euler vertex alone, and in that sense it is the first result here that uses Biot–Savart rather than translation invariance.

*Tao’s survivor, settled, with the sharp rate.* Corollary 4.9 closes the case for every inertia operator, isotropic or not, by the complementary route: his coupling needs one block to vanish identically, and no block of  $T_A$  vanishes for positive definite  $A$ . Corollary 4.22 supplies the rate  $\kappa^{-1}$ , which Proposition 4.11 shows is sharp; only the constant is lossy. Remark 4.2 records how an earlier and over-confident version of the statement went wrong before it could be proved.

Two features of the anisotropic case are worth stating as conclusions rather than as gaps, because in both the natural stronger statement is false and we know why. First, the distance formulation admits no uniform bound: by Remark 3.3 its infimum is zero, approached as  $A$  degenerates, in contrast with the uniform bound 0.51 of the isotropic case. The correct quantitative statement is the one in the joint spread, Corollary 4.22. Second, no blockwise rank argument can work for general  $A$ : by Remark 3.4 the map  $A \mapsto T_A$  is a submersion onto a single block at a general anisotropic point, so a single block *can* be rank one. This is why Theorem 3.2 proceeds by parity, and why the general case goes through the vanishing of a block rather than the rank of one.

One thing remains open.

*Sums, and what replaces them.* A *local* cascade operator is a finite linear combination of basic ones and can have rank greater than one on a block. Theorem 2.3 excludes each term separately; by Paper I’s Theorems 5.12 and 5.14 this does not by itself exclude the combination. Section 5 settles the question in the form in which it was open, and settles it the other way: sums are not excluded. Every Euler–Arnold vertex is a sum of at most eight tensors carrying one rank-one block each (Theorem 5.7), so no triad-local argument resting on the rank-one-per-block property can succeed (Corollary 5.8); and no linear argument can succeed either, the only linear functionals vanishing on the class being the energy identity, which every cascade operator satisfies by construction (Corollary 5.4).

What survives is a rigidity question across triads rather than on one. The tensors produced by Theorem 5.7 are manufactured from  $T_A$  and change from triad to triad, whereas a local cascade operator reuses one finite coefficient set at every scale and on every triad. Whether a single such set can agree with an Euler–Arnold vertex simultaneously on all triads of its support is open. For Tao’s own operator it cannot, by Corollary 4.9, whose hypothesis is a vanishing block rather than a rank condition — which is why this paper settles his four survivors notwithstanding Corollary 5.8.

## REFERENCES

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